Date : 17

Assignment:

Name : Vivek Kashyap
Class : SC
Subject : Mathematics
Roll no. : 65

Qus1. A sample of 35 cars has average milage 21.1 mpg

(a) Various miles per gallon
(b) type Quantitative variables (numerical)

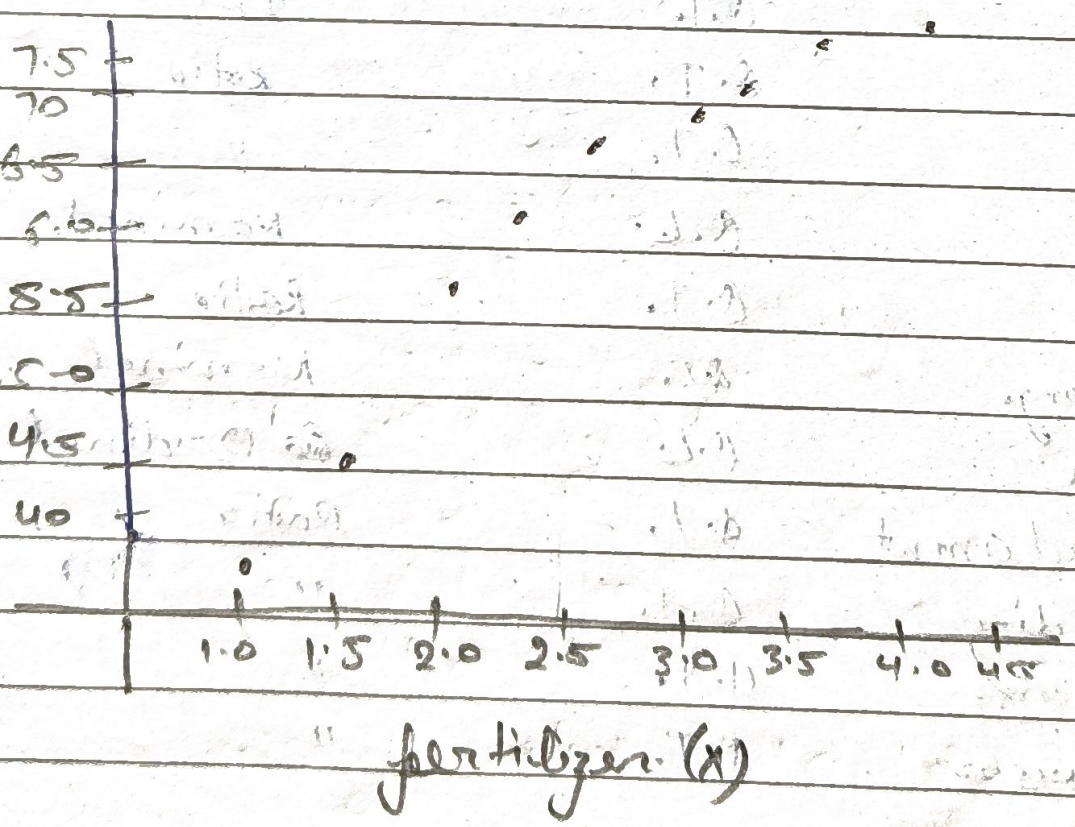
Qus2.

Variable	Type	Scale
Rating of movie	Qualitative	Ordinal
Birthday	Q.T.	Interval
Weight	Q.T.	Ratio
Height	Q.T.	"
Birthplace	Q.L.	Nominal
Total debt	Q.T.	Ratio
Stock exchange	QL.	Nominal
letter grades	Q.L.	Ordinal
credit card amnt	Q.T.	Ratio
Enter spending	Q.T.	"
Rainfall	Q.T.	"
No. of courses	Q.T.	"

Ques3. Frequency table

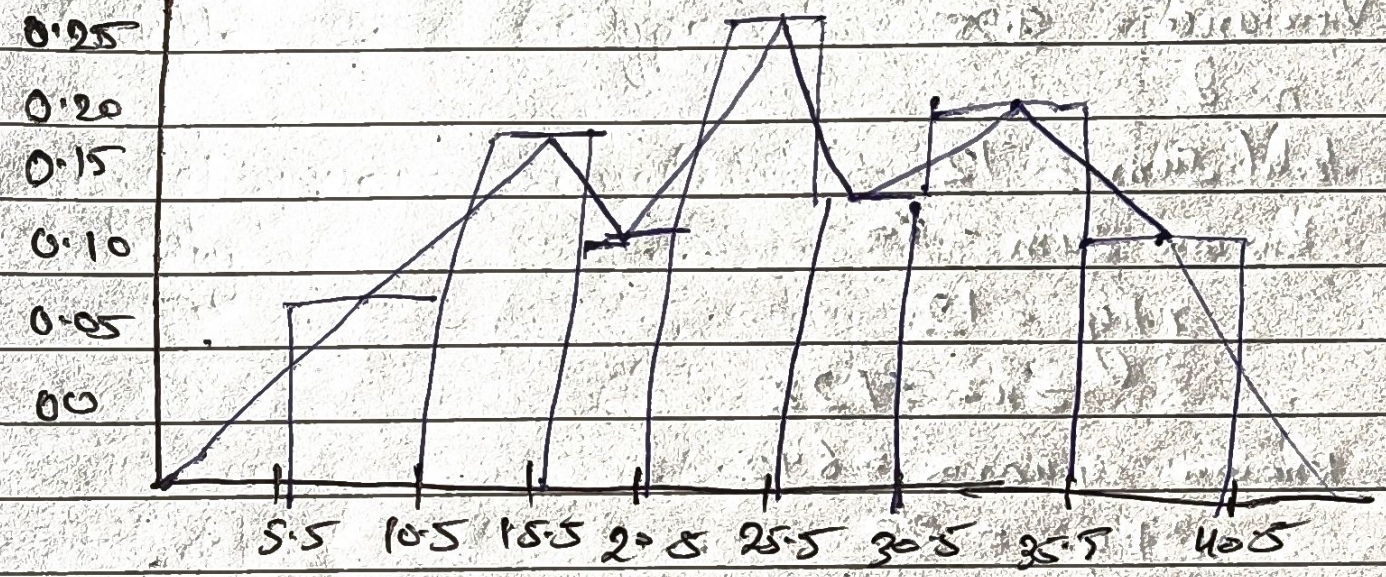
C.I.	Frequency (f)	(R.F.) (r.f)	(C.F.) (cf)
850 - 900	4	0.133	0.133
900 - 950	3	0.10	0.233
950 - 1000	5	0.167	0.400
1000 - 1050	4	0.133	0.533
1050 - 1100	5	0.167	0.700
1100 1150	6	0.20	0.900
1150 - 1200	3	0.10	1.000

Ques4. Scatter Diagram



Q5.

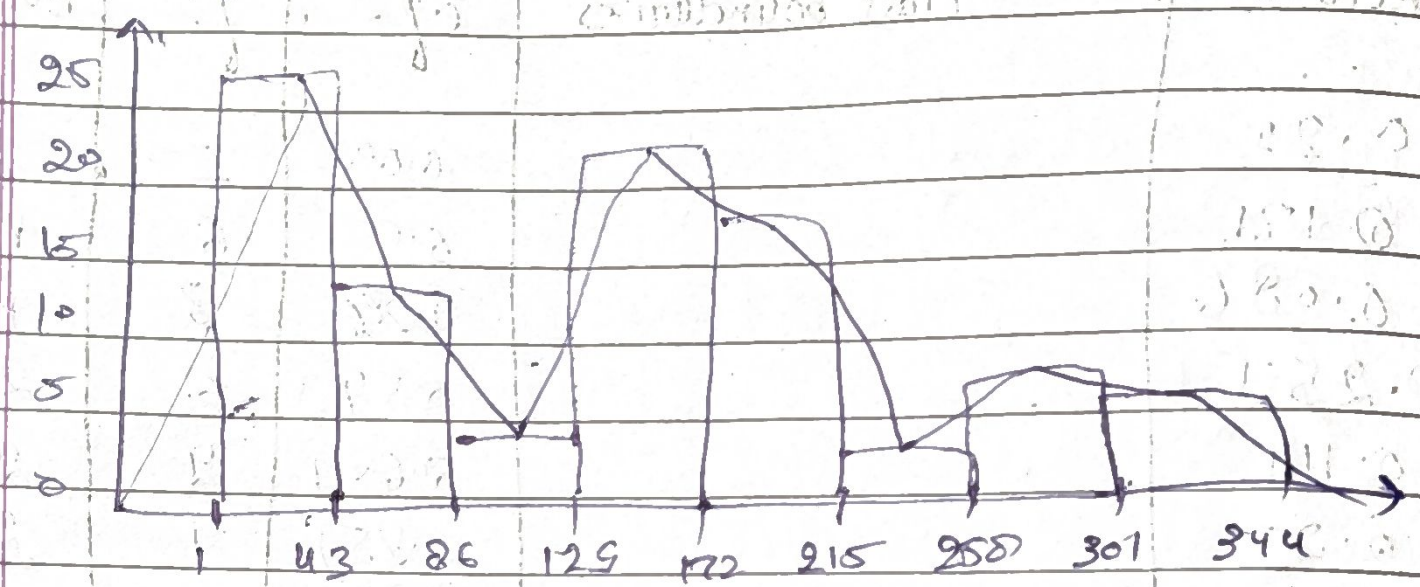
Relative	Class boundaries	c.f.	f.	Midpoint
0.29		0.09	1	2
0.171		0.2	6	13
0.086		0.286	3	18
0.257		0.543	9	23
0.114		0.657	4	28
0.200		0.857	7	23
0.143		1	5	30



Q6.

C.I.	(f)	Relative f. (rf)	Midpoint	c.f.
1-43	24	0.381	22	24
44-86	12	0.190	65	36
87-129	3	0.048	108	39
130-172	14	0.222	151	53
173-215	1	0.016	194	54
216-258	2	0.032	237	56
259-301	4	0.063	280	60
302-344	3	0.048	323	63

Histogram



Q7. Mean = 9

Variance = 9.25

n = 8

total sum = 72

known sum = 60

$x + y = 12$

$\Sigma x^2 = 722$

known squares = 642

$x^2 + y^2 = 80$

$(x+y)^2 = x^2 + y^2 + 2xy$

$144 = 80 + 2xy$

$xy = 32$

$x^2 = 12x + 32 = 0$

$x = 4, 8$

missing values = 4 and 8

Ans. from the given graph

(a) Max ratio (investement/profit) for A is at highest peak of company A curve

(b) min investment for B at lowest point of company B wave.

Q9. Data (weight in kg)

48, 53, 60, 68, 72, 77, 78, 80

